

PHYS 1401 COLLEGE PHYSICS

REVIEW FOR EXAM 4 CH 10-12

Familiarize yourself with the formula sheet provided in the course resources. It may look a little different to the way you've been working with the equations. The equations can be represented in many ways, and personal preference is often used by physicists.

Review Questions

1. Define thermal contact
2. Define thermal equilibrium
3. Define heat for a system (Q). When is it positive? Negative? How is it different from the internal energy of a substance?
4. Define the methods of heat transfer (conduction, convection, radiation) and be able to tell the differences between them. What are some examples of each type of transfer?
5. State the relationship between linear, area, and volume expansion coefficients.
6. State the equation and define the first law of thermodynamics. For W , work done, when is it positive? When is it negative? When is ΔU positive? Negative?
7. What is the difference between isobaric, adiabatic, isovolumetric, and isothermal processes?
8. State the second law of thermodynamics. What does it tell us about the amount of heat that an engine can convert into work? What does it tell us about the efficiency of a heat engine?
9. The burning of fossil fuels produces greenhouse gases. What are two other sources of greenhouse gas emissions?
10. State the equation of entropy. What is the only type of process it applies to?
11. A temperature change from 25°C to 40°C corresponds to what incremental change in F ?
Ans: 27°F

12. Convert 289.62K to °C.

Ans: 16.47°C

13. Find the coefficient of linear expansion for aluminum. If a bar of aluminum of length L is heated from 20°C to 65°C and the change in length is measured to be 0.6m, what was the initial length L of the aluminum?

Ans: 556m

14. How many moles of Helium gas are there if it is held at 27°C and has an internal energy of 9.5×10^4 J?

Ans: 25 moles

15. Consider an ideal gas. If the pressure is doubled while the amount of gas and temperature is kept constant, what happens to the volume?

Ans: changes by a factor of 1/2

16. 50g of ice is heated from -10°C to 0°C. How much heat was required to increase this temperature? The ice then fully melts. How much heat was required for the phase change? $L_f = 3.33 \times 10^5$ J/kg, $c_{ice} = 2090$ J/kg°C

Ans: 1045 J, 1.67×10^4 J

17. An Iron bar conducts energy across its length of 35cm. What is the rate of heat flow if one end is at 10°C and the other end is at 22°C? The bar has a cross sectional area of 80cm². How much heat is transferred in 30 seconds? Iron thermal conductivity $k = 79.5$ W/m°C

Ans: 21.8 W, 654 J

18. Neon gas undergoes adiabatic expansion from 100m³ to 200m³. Neon has an adiabatic index of 1.64. If the initial pressure is 2.1×10^5 Pa, what is the final pressure?

Ans: 6.7×10^4 Pa

19. In an isobaric process, there is a volume change of 10m³. If 2×10^5 J of work is done on the gas to cause this change, what was the pressure of the gas?

Ans: 2×10^4 Pa

20. A heat engine operating between a pair of hot and cold reservoirs with respective temperatures of 400K and 100K will have what maximum efficiency?

Ans: 75%

21. Systems A and B are placed in thermal contact. If the temperature of system A is larger than the temperature in system B, what will happen to the entropy of system A?

Ans: it decreases

22. A certain process involves a heat transfer of 500J and results in a change in entropy of $\Delta S = 1.25 \text{ J/K}$. Calculate the temperature T at which this process occurs.

Ans: 400K